Sophie Germain — Recherches sur la théorie des surfaces élastiques (1821)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Sophie Germain, who entered a prize competition as its only entrant, returned after an honourable mention, and won it under her own name. She had written to Lagrange and to Gauss as Monsieur Antoine-Auguste Le Blanc. When her concern for Gauss's safety forced her to reveal who she was, his praise, on the record, increased. In 1821 she published her prize memoir at her own expense, to answer Poisson in public rather than let the argument rest.

The author's own terms

A note on "elastic figure" and "natural figure": in § I, article 1, following the definition she quotes from d'Alembert's *Encyclopédie*, Germain calls the elastic figure the figure an exterior cause has forced a surface to take, and the natural figure the figure it had before that action.

The story

Marie-Sophie Germain (1776–1831) was born on 1 April 1776 in Paris, the middle of three sisters. Barred as a woman from the newly opened École Polytechnique, she obtained its lecture notes and submitted work to Joseph-

Louis Lagrange under the name Le Blanc. Lagrange requested a meeting, learned who she was, and became her mentor. From 1804 she corresponded with Carl Friedrich Gauss under the same name. Her identity was revealed after she intervened, through General Pernety, for Gauss's safety during the French occupation of Brunswick, and Gauss replied with documented admiration.

She never married. Because of prejudice against her sex, she could not make a career of mathematics, and worked independently, supported by her family's means. Breast cancer was diagnosed in 1829; she worked through it, and died at 13 rue de Savoie, Paris, on 27 June 1831, aged fifty-five. Her death certificate listed her not as a mathematician but as a *rentière*, an annuitant or property holder.

The work and what survives

The *Recherches* grew out of the Paris Academy of Sciences prize competition on Ernst Chladni's vibrating-plate experiments. Denis Poisson became a judge rather than a contestant, leaving Germain the only entrant. Her first submission, in autumn 1811, was not crowned; the commission held that the true equations were not established, though Lagrange derived an equation from her hypothesis. Her second, in 1813, received an honourable mention only. With her third, under her own name, she won on 8 January 1816, the first woman to win a prize from the Paris Academy of Sciences.

She published the prize essay at her own expense in 1821, mostly to present her work in opposition to Poisson's, pointing out errors in his method. The book runs to x + 96 pages with one fold-out plate (Paris: Huzard-Courcier, 1821). Contemporary reception was mixed, polite to indifferent, but Claude-Louis Navier's reported verdict survives: a work few men could read and only one woman could write. The text survives in print and digitised form; a Cambridge Library Collection reprint reproduces the French, not a translation.

Its English fortunes

The memoir's ideas are discussed wherever elasticity is taught, and the secondary literature treats them in English. That discussion is not a translation of the book itself.

The finding, at its true strength, is this: **no English translation of this memoir was found after research.** The searches behind that sentence covered the 1821 title with "English translation," Germain's biography and publication history, and the Cambridge reissue. They did not include an exhaustive catalogue search for partial English renderings of excerpts, and the wording must stay within that bound.

What completing it would take

The translation prepared for this collection is partial. It gives the Avertissement, printed pages v–x, and § I, printed pages 1–14, articles 1–8, complete, in a fresh English translation from the French of 1821. Section II begins on printed page 15 and is not translated there.

Completing it would mean translating §§ II–IV, printed pages 15–96, with the Errata and the fold-out plate. The whole would then need collation against page images of the printed original, with every reconstructed formula re-verified, since the machine-read text is reliable in prose and treacherous in formulas. It would also need the two reviews set for it before any publication: a historian of mathematical physics, and a reader of the period French. The notes apparatus referred to in the edition note, with its leaf-to-printed-page map, would need to be recovered or rebuilt. Her errors, which she admits in the Avertissement, are to stay annotated, not corrected.

Sources

- https://en.wikipedia.org/wiki/Sophie_Germain (life, correspondence, prize history)
- https://news.cnrs.fr/articles/rebel-with-a-cause (École Polytechnique; Lagrange)
- https://old.maa.org/node/2592666 (the 1821 memoir)
- https://www.barnesandnoble.com/w/recherches-sur-la-theorie-des-surfaces-elastiques-sophie-germain/1110500527 (Cambridge French reprint)